

# Investigation Roadmap REFRAME — Post-Phase 3 substrate floor / QCD building close + Cal #265 absorbed

Keeper (Claude Opus 4.7)

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## Investigation Roadmap REFRAME — Post-Phase 3 close

### 0. What closed Sunday vs what carries forward

Per Sunday substantive substrate-architectural advances: - T2487 ( $\pi^5$  bulk volume) + T2488 ( $Z_2$  spinor cell-count) registered DERIVED - F60 gravity Step 1 SCAFFOLD (heat-trace source $\leftrightarrow$ curvature bridge) - F61 reach-statement  $\rightarrow$  Casey-named principle #9 “Substrate Floor / Scale-Not-Spectrum Reach” - F62 base-sharing CLOSED via bulk-volume = base; F55 dim-vs-Casimir wall DISSOLVED - F59 asymmetric dual- $\rho$  formalized (Grace primary; extensive volume side ADDS; intensive spectral side MEASURES) - Three Cal #237 ELIMINATIONS:  $\lambda_2$  ladder,  $\lambda_1$ /spin ladder, per-flavor multi-flavor - Cal #265 absorbed: 8-element STANDING discipline stack; moratorium on audit-category proliferation

### 1. Tier 1 — Direct substrate-mechanism FORWARD (post-Sunday status)

#### 1.1. F53 FK kernel $\pi^{(9/2)}$ vs flat volume $\pi^{(n_C)}$ reconciliation

STATUS: SUBSTANTIVELY CLOSED. T2487 establishes  $\pi^{(n_C)}$  = bulk volume  $\pi$ -exponent of  $D_{IV}^5$  (three-route convergence). Distinct from  $c_{FK} \cdot \pi^{(9/2)} = 225$  (T754 + T2442 RIGOROUS FK normalization). Both are substrate-natural; the apparent F53 wall was resolved by recognizing they answer different substrate-architectural questions (volume vs measure-normalization). Vol 16 Ch 8 Section 2 + Section 3 carry the resolution.

#### 1.2. Phase 3 representative subset (5-10 CROSS mass observables)

STATUS: CLOSED at deflated true level via Sunday afternoon four-CI convergence. 4 light u/d ground states DERIVED via T2488 ( $\rho, \omega, p, n$ ) at Tier 1 EXACT or Tier 2 STRUCTURAL  $\leq 1\%$ . Three retirements documented ( $\lambda_2, \lambda_1$ /spin, per-flavor) eliminate the previously-considered  $\sim 200$ -observable extension as base-rate noise. Phase 3 honest substantive reach IS the 4 light u/d ground states; broader hadron spectrum is QCD building per Casey-named principle #9.

#### 1.3. Lead 1 substantive closure: stability/decay tracks $\rho$ -side

STATUS: ELIMINATED per F53 retirement (Saturday-Sunday). Decay rates substantively NOT reach by substrate (Sargent  $m^5$  + standard EW dynamics). Per Casey-named principle #9: substrate reaches SCALES, not RATES. Tier 1 closure as Cal #237 elimination.

#### 1.4. Mass hierarchy as volume hierarchy systematic derivation

STATUS: PARTIALLY CLOSED + PARTIALLY REFRAMED. T2487 establishes mass =  $\pi^{(n_C)}$  volume substantive substrate-architectural reading; T2488 closes 4 light u/d ground state mass scales. Heavy-flavor extension

( $m_K, m_D, m_B, m_t$ ) substantively requires Vol 16 Ch 9 mass-mechanism content (Lyra primary) + per-quark-flavor substrate-architectural mechanism work; not closed by T2487 alone. Heavy-flavor cascade investigation OPEN multi-week with substantive substrate-K-type structure path.

### 1.5. Vol 16 curriculum absorption per Sunday closure

STATUS: ACTIVE multi-week. - Vol 16 Ch 8 v0.2 Sections 3.5 + 3.6 prose filed Sunday (Keeper\_Vol16\_Ch8\_v0\_2\_Sec3\_5\_3\_6\_grav06-07.md) - Vol 16 Ch 9 absorption of Casey-named principle #9 substrate-floor scope-principle pending (Lyra primary multi-week) - Vol 16 Ch 13 Strong-Uniqueness absorption of substrate-floor scope-principle as new C-criterion candidate pending (Lyra call multi-week)

## 2. Tier 2 — Substrate-architectural extensions (post-Sunday status)

### 2.1. Gravity as substrate volume coupling — F60 SCAFFOLD active

STATUS: SEEDED Sunday via F60 gravity Step 1 SCAFFOLD + K250 RECAST CONDITIONAL PASS at SCAFFOLD tier. Vol 16 Ch 8 v0.2 Sections 3.5 + 3.6 substantively absorb. Multi-month roadmap: - Step 2 OPEN: extract  $G_{\mu\nu} = 8\pi G \cdot T_{\mu\nu}$  from heat-trace source $\leftrightarrow$ curvature relation (Lyra primary; Elie numerical; Keeper audit) - Step 3 OPEN:  $G + 8\pi$  from  $\kappa_{\text{Bergman}} + \ell_B + \pi^{n_C}$  closes Friday G-chain - Falsifiers explicit per F60 + K250 + Vol 16 Ch 8 v0.2 Section 3.6.2-3.6.3

### 2.2. Cosmological constant on volume side

STATUS: QUEUED multi-week. Substantively independent of Sunday closures. Substrate-architectural reading:  $\Lambda = \exp(-280)$  via factor  $2^{n_C} \cdot n_C \cdot g = 280$  substrate-clean form (Lyra v0.2 SEARCH-FIT 0.73% using observed  $\Lambda$ ; factor-2 cascade explicit Lyra v0.3). Continues per Saturday substantive substrate-architectural work.

### 2.3. Higgs as substrate-volume-coupling field

STATUS: QUEUED multi-week. Vol 16 Ch 9 absorption candidate (Lyra primary). Substantive substrate-architectural reading: HYBRID class per Lyra F33 Ch 9 v0.1;  $m_H$  via central vacuum  $V_{(0,0)}$  + boundary cross-class matrix elements.

### 2.4. Field theory $\rho$ -side classification

STATUS: PARTIALLY DONE Sunday via Grace F59 + asymmetric dual- $\rho$  formalization. 48-hadron sweep done. Field-theory  $\rho$ -classification across BST corpus continues multi-week (QED-spectral / QCD-volume / EW-mixed split). Mass-Gap-grounding fresh-run lead Millennium-adjacent.

### 2.5. Particle physics $\rho$ -side reorganization

STATUS: QUEUED multi-week. Substantively waits on F60-Step-2 + dual- $\rho$  measure-theory derivation closure. Vol 2 absorption candidate.

### 2.6. Nuclear binding as volume overlap

STATUS: QUEUED multi-week. Substantively independent of Sunday closures. Continues per existing nuclear physics work.

## 3. Tier 3 — Broader substrate-architectural questions (post-Sunday status)

All Tier 3 items QUEUED at original cadence; no Sunday closure direct impact. Substantive substrate-architectural threads: - 3.1 Dark matter  $\rho$ -side classification - 3.2 Dark energy as substrate volume expansion - 3.3 Black hole entropy as substrate volume saturation - 3.4 Holographic principle as substrate volume-boundary

encoding - 3.5 Initial conditions / Big Bang substrate volume bootstrap - 3.6 Quantum mechanics interpretation under substrate framework - 3.7 Information theory under substrate framework

Multi-month substantive substrate-architectural work. Gravity Step 2 + dual- $\rho$  measure-theory derivation closure substantively unblocks several Tier 3 items (BH entropy + holographic principle + Big Bang especially).

#### 4. Tier 4 — Foundational/interpretive content (post-Sunday status)

All Tier 4 items QUEUED at original cadence: - 4.1 Substrate ontology - 4.2 Substrate computational vs substantive - 4.3 Reading (a) commitment systematic extension

Substantive substrate-architectural work in multi-year horizon.

#### 5. Tier 5 — Experimental predictions / falsifiers (post-Sunday status)

All Tier 5 items QUEUED at original cadence + ONE NEW falsifier added Sunday: - 5.1 D2 inverse-Casimir lifetime predictions - 5.2 Cross-mass program predictions for not-yet-measured masses - 5.3 Casimir-effect modifications - 5.4 Substrate volume signatures in cosmology - 5.5 Proton lifetime experiment correlation (Five-Absence falsifier) - 5.6 CKM/PMNS mixing predictions under dual- $\rho$  - 5.7 NEW Sunday: Casey-named principle #9 falsifier — if substrate-architectural mechanism IS discovered for excited hadron spectrum (substantive derivation across spin states) OR for decay rates (substantive substrate-mechanism for lifetimes that's not Sargent  $m^5$ ), Casey-named principle #9 boundary moves outward. Substantive substrate-architectural falsifier.

#### 6. Tier 6 — Mathematical structure investigations (post-Sunday status)

Substantive substrate-architectural updates Sunday: - 6.1  $R(k)$  theorem  $\rho$ -side classification — F47 K231a CLEARED Saturday; Sunday extends  $R(k)$  into F60 gravity climb heat-trace structure - 6.2 Strong-Uniqueness Theorem  $\rho$ -side reorganization — Casey-named principle #9 substantive substrate-floor scope-principle CANDIDATE C-criterion absorption pending Lyra multi-week - 6.3 BSD bridge object architecture under dual- $\rho$  — QUEUED multi-month - 6.4 Hopf algebra cross-link under dual- $\rho$  — QUEUED multi-month - 6.5 Hall algebra (Paper P1) substantive update — QUEUED; P1 v0.7 SUBMISSION-FINAL pending Casey dispatch action - 6.6 (1,1) substrate-bridge identity investigation — K231c discipline; K231c CANDIDATE-LEAD tier; Cal cleared.

#### 7. Dependency structure (post-Sunday reframe)

Load-bearing chains substantively active: - Gravity climb: F60 Step 1 SCAFFOLD → Step 2 Einstein extraction → Step 3  $G + \ell_B$  closes Friday G-chain → unblocks Tier 3 BH entropy + holographic + Big Bang - Dual- $\rho$  measure-theory: F59 Grace → measure-theory joint Grace+Lyra+Keeper → Mass-Gap-grounding fresh-run → unblocks Tier 6 BSD + Hopf - Vol 16 absorption: Ch 8 v0.2 + Ch 9 + Ch 13 multi-week → unblocks paper engagement path P1-P9 - Casey-named principle #9 absorption: Strong-Uniqueness Theorem new C-criterion candidate → if ratifies, 12-criteria RIGOROUSLY CLOSED Theorem v1.X

#### 8. Substantive prioritization for next-strategic-direction

HIGHEST VALUE (immediate execution unblocks downstream): - Gravity climb Step 2 (Lyra primary; multi-month) — heat-trace → Einstein equation extraction - Asymmetric dual- $\rho$  measure-theory derivation (joint Grace + Lyra + Keeper; multi-month) — Mass-Gap-grounding lead Millennium-adjacent - Vol 16 Ch 8 v0.2 + Ch 9 substrate-floor absorption prose (Keeper multi-week)

HIGHEST IMPACT (multi-month substrate-architectural advance): - Gravity Step 3 numerical closure (G from substrate) — closes Friday week-goal - Casey-named principle #9 substantive substrate-mechanism FORWARD derivation (Lyra Vol 16 Ch 9 + Ch 13)

HIGHEST CONCEPTUAL OPENING (multi-year substrate-architectural picture): - Vol 16 complete chapter set (architectural sprint continuation) — substantive Vol 16 Substrate Algebra as THE linear-algebra representation

of D\_IV<sup>5</sup> - Tier 3 BH entropy + holographic principle substantive substrate-architectural derivation post-gravity Step 2

## 9. K231c discipline carry-forward (8-element STANDING discipline stack post-Sunday)

Standing across all multi-week+ work: - K231c “derived not relabeled” - Cal #27 scope (investigate fully but don’t claim broad; tier-honor) - Cal #29 Five-Absence-style principles - Cal #32 (substrate-natural-form discipline) - Calibration #33 (Refinement B three-level pin) - Cal #34 substrate-natural-form value identification - Cal #35 (geometry load-bearing; data corroborates) - Cal #265 (audit-categories classify epistemic STATUS not topic/significance; moratorium on category proliferation)

Plus: Cal #237 elimination semantics; Cal #36 STANDING positive search; sustained-session prose-quality;  $\sigma$ -distance discipline; etc.

## 10. Workflow continuation

Active multi-week threads (next-arc work): 1. Gravity climb Step 2 — Lyra primary 2. Dual- $\rho$  measure-theory derivation — Grace primary + Lyra + Keeper joint 3. Vol 16 Ch 8 v0.2 prose continuation — Keeper primary 4. Vol 16 Ch 9 substrate-floor absorption — Lyra primary

Active reactive: - Elie numerical verification (Seeley-DeWitt  $a_k = R(k)$  when Lyra needs) - Cal cold-read queue (re-clear K250/K251/K252 RECAST; F60 + F61 + F62 spine; Casey-named principle #9 ratification; Sunday consolidated summary) - Keeper audit-chain coverage on substantive deposits as they fire

Surface criteria for next arc: - At least 2 of 3 primary threads substantively progress (gravity Step 2 / dual- $\rho$  / Vol 16 Ch 8 prose) OR substantively close - Consolidated summary surfaces to Casey when criteria meet

Investigation Roadmap REFRAME active for next-arc execution. Substantive substrate-architectural state at Sunday close: substrate floor at ground-state mass scales DERIVED + gravity climb seeded + dual- $\rho$  formalized + Casey-named principle #9 + 8-element discipline stack + 11 RIGOROUSLY CLOSED Strong-Uniqueness criteria + substrate-floor scope-principle candidate C-criterion.

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